GGG-F-001391
October 30, 1967

INTERIM FEDERAL SPECIFICATION

FILER, PISTON RING

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D.C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 This specification covers a portable, rotary piston ring filer for sizing various piston rings and slotting pistons skirts.

1.1.1 Federal specification coverage. Federal specifications do not include all the types and styles indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424 - Hand Tools; Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5110

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:43Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Governmental:

National Bureau of Standards (NBS) Handbook:
H28 - Screw-Thread Standards for Federal Services, and Supplements thereto.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Illustration. The illustration shown herein is for the convenience of identification, and is not intended to preclude the purchase of piston ring filers which are otherwise in accordance with this specification.

3.2 Design. The filer shall be of standard commercial design. All parts subject to wear, breakage, or distortion shall be accessible for adjustment and repair.

3.3 Materials. Materials shall be as specified herein. Materials not specifically designated shall be of good commercial quality.

3.4 Construction. The filer shall be complete so that it can be used for the operations specified herein.

3.4.1 Castings. All castings shall be of uniform quality, free from blowholes, porosity, hard spots, shrinkage defects, cracks, and other injurious defects that may directly or indirectly affect the structural strength or the serviceability of the castings involved. Strength and other essential physical and chemical properties of the castings shall be adequate throughout for the intended purpose.

3.4.2 Welding, brazing and soldering. Welding, brazing and soldering shall be performed in accordance with good commercial practice. In no event shall such processes, including peening or plugging, be used on castings or forgings for reclaiming any parts of the filing machine without authorization of the procuring activity.

3.4.3 Fastening devices. Parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 Threads. All threaded parts shall be in the inch system and shall comply with H28.

3.4.5 Interchangeability. All replaceable parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model can be replaced or adjusted without requiring modification.

3.4.6 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be clean and and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

3.5 Performance and product characteristics. The piston ring filer shall be capable of providing smooth surfaces parallel within 0.002 inches for piston ring clearances. This tool shall also be capable of producing from a 90° square cut, to a 45° left cut, and a 45° right cut.

3.5.1 Weight. Total weight of the assembled piston ring filer shall be no greater than 3 pounds.

3.6 Detail of components. The filer shall consist of the following principal component parts:

(a) Housing
(b) Handcrank
(c) Gage plate
(d) Cutter file
(e) Mounting bracket

3.6.1 Housing. The housing of the piston ring filer shall be of metal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

3.6.2 Handcrank. A handcrank shall be furnished to rotate the shaft on which the cutter file is mounted. Turning the handcrank shall provide a continuous rotary cutting stroke of the file.

3.6.3 Gage plate. A calibrated gage plate shall provide cutter file and piston ring alignment when the ring is held against the plate. Guide pins shall be located in the gage plate to support the piston ring ends against the cutter file. The gage plate shall be calibrated 45° left of center, center, and 45° right of center.

3.6.4 Cutter file. The cutter file shall be of tempered tool steel, and milled across each face for filing. The edges of the cutter file shall be designed with cutting teeth around the periphery for use as a rotary cutter.

3.6.5 Mounting bracket. A mounting bracket shall be provided for securing the filer to an engine head stud or for bolting the filer to a work bench.

3.6.6 The shape of the piston ring filer shall be similar to figure 1.

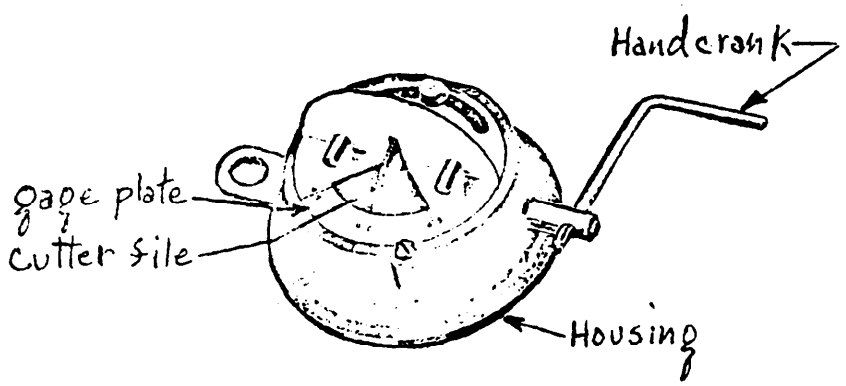


FIGURE 1. Piston ring filer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

TABLE I. Sampling data

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms Of	Reference
Visual examination	1 each	II	1.5	Defects per hundred units	4.3.1
Dimensional and weight examination	1 each	S-4	1.5	do.	4.3.2
Testing	1 each	S-3	2.5	do.	4.4
Preparation for delivery	one container	S-2	4.0	do.	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table II.

TABLE II. Classification of defects

Categories	Defects
Critical:	
1	None defined.
Major:	
100	Design not as specified (see 3.2).
101	Construction not as specified (see 3.4).
102	Defective casting (see 3.4.1 and 3.4.2).
103	Fastening devices unsatisfactory (see 3.4.3).
104	Threads not in compliance with H28 (see 3.4.4).
105	Component parts not as specified (see 3.6).
106	Faulty workmanship (see 3.10).
Minor:	
200	Interchangeability not as specified (see 3.4.5).
201	Surfaces unsatisfactory (see 3.4.6).
202	Weight not as specified (see 3.5.1).
203	Spare parts not as specified (see 3.7).
204	Finish inadequate (see 3.8).
205	Sheet of instructions not as specified (see 3.9).
206	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.

4.3.2 Dimensional and weight examination. Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.4 Test methods.

4.4.1 Test procedures. Each sample unit shall be tested in accordance with 4.4.2 thru 4.4.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

4.4.2 Performance. The filer shall be tested for compliance with 3.5. A new piston ring shall be inserted in the filer and filed until both faces of the ring gap have been resurfaced. Remove and set the filed piston ring on a flat surface with the desired gap set by the proper feeler gage, and the ends of the piston ring held closed on the gage. If it is possible to insert a 0.002 inch thick feeler gage between the ends of the piston ring at any point, the filer shall be rejected.

4.4.2.1 Cutter file. The ring shall be placed again on the filer and refiled with fifty revolutions of the handcrank. The cutter file used in the performance test shall not show undue wear.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table III.

TABLE III. Classification of defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Contents	Quantities not as specified.
Containers	Not the kind specified.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate strapping; distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. The piston ring filers shall be preserved, packaged, and packed in accordance with MIL-H-15424. The levels of preservation and packaging shall be A or C and the level of packing A, B, or C, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The primary intended use of the filing machine specified herein is for piston ring gap filing when fitting piston rings.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation, packaging, and packing required (see 5.1 and 5.2).

GSA DC 68-4607

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.